Multimodal Intent Recognition

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***Abstract*—Intent recognition aims to identify the underlying intention of a speaker by jointly analyzing textual, acoustic, and visual information. Compared to unimodal approaches, multimodal intent recognition provides richer contextual understanding by exploiting complementary information across modalities. Recently, Transformer-based multimodal architectures have demonstrated significant improvements by incorporating modality alignment, prompt learning, and contrastive representation learning. In this project, we reproduce the Token-Level Contrastive Learning with Modality-Aware Prompting (TCL-MAP) framework [2] for multimodal intent recognition on the MIntRec benchmark dataset [4]. The reproduction involves setting up the complete training pipeline, preparing the multimodal features, understanding the architecture, and validating the reported results through experimental evaluation. We further situate TCL-MAP against the Modality-Invariant and -Specific Representations (MISA) framework [3], using the head-to-head comparison reported by Zhou et al. [2] on the MIntRec and MELD-DA [5] datasets, and additionally examine the reproduced model's behavior when extended to the IntentHQ dataset introduced by Atreya et al. [1]. Experimental results confirm the effectiveness of multimodal, prompt-based contrastive learning for intent recognition and provide insights into the relative strengths of prompt-based fusion (TCL-MAP) versus modality-invariant/-specific representation learning (MISA). This work contributes a reproduction study together with a literature-grounded comparative analysis of two representative multimodal intent recognition frameworks, situated alongside a related intent-aware video analysis pipeline.**

I. Introduction

Intent recognition is a fundamental task in human-computer interaction that aims to identify the underlying intention expressed by a user. Traditional intent recognition systems primarily rely on textual information; however, human communication naturally involves multiple modalities such as speech, facial expressions, gestures, and vocal characteristics. These complementary modalities often contain important contextual cues that cannot be captured from text alone.

Recent advances in deep learning and Transformer-based architectures have significantly improved multimodal representation learning by jointly modeling textual, acoustic, and visual information. Multimodal Intent Recognition (MIR) has attracted considerable attention because of its applications in intelligent virtual assistants, conversational AI, customer service systems, healthcare assistants, educational platforms, and human-robot interaction.

TCL-MAP [2] introduces a multimodal architecture that combines similarity-based modality alignment, modality-aware prompt generation, MAG-BERT-style fusion, and token-level supervised contrastive learning to improve intent recognition performance. Rather than simply concatenating multimodal features, TCL-MAP learns aligned semantic representations and injects modality-specific prompts into a BERT encoder, enabling richer interaction among modalities.

MISA [3] approaches multimodal learning from a different perspective. Instead of prompt learning, MISA explicitly separates modality-invariant information shared across modalities from modality-specific information unique to each modality, using similarity, difference (orthogonality), and reconstruction losses, followed by Transformer-based self-attention fusion.

The primary objective of this project is to reproduce the official TCL-MAP implementation on MIntRec and validate its experimental performance. We then compare TCL-MAP against MISA using the benchmark numbers reported in the TCL-MAP paper itself [2], which evaluates both methods under identical splits on MIntRec and MELD-DA. Finally, to broaden the scope of the study, we examine how the reproduced TCL-MAP pipeline behaves when applied to IntentHQ [1], a recent dataset for fine-grained, human-centric intent recognition in video.

II. Motivation

Reproducing state-of-the-art multimodal research is challenging due to complex training pipelines, multimodal feature preprocessing, and limited implementation detail in publications. Reproducibility plays an essential role in validating published research and establishing reliable baselines for future work.

TCL-MAP introduces several interacting components: similarity-based modality alignment, modality-aware prompt learning, multimodal fusion, and token-level contrastive learning. Understanding how these components interact requires both theoretical analysis and hands-on implementation. In parallel, MISA represents a structurally different paradigm, factorizing each modality into invariant and specific subspaces rather than using prompt-based fusion. Comparing these two architectures on a shared benchmark helps identify their relative advantages and limitations for multimodal intent recognition.

This project therefore aims to (i) reproduce the official TCL-MAP implementation, (ii) analyze its architecture and training pipeline, (iii) compare it against MISA using the paper-reported benchmark results on MIntRec and MELD-DA, and (iv) explore the pipeline's generalization by applying

it to the IntentHQ dataset, situating it against a conceptually related, more recent intent-aware video analysis framework.

III. Literature Review

Multimodal intent recognition has become an active research area in natural language understanding by integrating textual, acoustic, and visual information to infer user intentions more accurately. Unlike traditional text-based approaches, multimodal methods exploit complementary information from speech, facial expressions, and language, resulting in more robust intent prediction in real-world conversational systems.

Early multimodal learning approaches mainly focused on feature fusion. Tensor Fusion Network (TFN), Low-rank Multimodal Fusion (LMF), and Memory Fusion Network (MFN) combined information from different modalities using tensor operations and memory mechanisms. Although these methods improved multimodal representation learning, they suffered from high computational complexity and were limited in capturing long-range cross-modal dependencies.

The introduction of Transformer architectures significantly improved multimodal learning. Models such as MulT employed cross-modal attention to capture interactions between text, audio, and visual modalities, while MAG-BERT incorporated multimodal information into the BERT encoder through a Multimodal Adaptation Gate. These methods demonstrated superior contextual modeling but still faced challenges in reducing the semantic gap among heterogeneous modalities.

MISA [3] addressed this limitation by explicitly decomposing each modality into modality-invariant and modality-specific representations. The invariant representations, learned with a Central Moment Discrepancy (CMD) similarity loss, capture information shared across modalities, while the specific representations, constrained by an orthogonality (difference) loss, preserve unique modality characteristics; a reconstruction loss prevents the specific encoders from collapsing to trivial solutions. The six resulting vectors are fused via Transformer self-attention and passed to a classifier. MISA was originally proposed for sentiment analysis and humor detection, and was evaluated on the CMU-MOSI, CMU-MOSEI, and UR_FUNNY benchmarks [3], where it outperformed prior state-of-the-art models such as ICCN and MulT.

More recently, TCL-MAP [2] introduced a prompt-based multimodal framework specifically for intent recognition. Instead of directly concatenating multimodal features, TCL-MAP first performs similarity-based modality alignment (SBMA) to reduce the distribution gap between modalities, then generates modality-aware prompts via a cross-modality attention module (MAP), and concatenates these prompts with the BERT text tokens. Token-level contrastive learning (TCL) further constructs augmented sample pairs – one ending in a `[MASK]` token and one ending in the ground-truth label token – and applies an NT-Xent loss between their encoded representations, in addition to a standard cross-entropy classification loss. TCL-MAP was evaluated on MIntRec [4] and MELD-DA [5], where the authors directly compare it against MAG-BERT, MulT, and MISA under an identical experimental protocol, reporting consistent improvements over all three baselines [2].

In a related but distinct direction, "Beyond Real versus Fake: Towards Intent-Aware Video Analysis" [1] proposes an intent-oriented framework built on IntentHQ, a new dataset of 5,168 manually curated and annotated YouTube videos spanning 23 fine-grained intent categories grouped into five broad classes: Deception, Emotional Engagement, Information Delivery, Persuasion, and Reputation. Rather than binary real-versus-fake classification, this framework targets high-level communicative intent. Its strongest configuration is a self-supervised, three-way contrastive alignment scheme that pretrains modality-specific encoders (CLIP ViT-L/14 for video, WavLM for audio, and the CLIP text encoder for transcripts) by jointly aligning all three pairwise modality combinations (text-video, video-audio, audio-text) via an InfoNCE-style loss, followed by supervised fine-tuning of a lightweight MLP classifier. This pipeline achieves 52.5% accuracy on the 23-way IntentHQ task, compared to 40.3%/44.3% for purely supervised MLP/cross-attention baselines using frozen pretrained encoders [1].

Although MISA, TCL-MAP, and the IntentHQ framework all aim to improve multimodal representation learning, they adopt fundamentally different strategies: MISA factorizes modalities into invariant/specific subspaces before fusion; TCL-MAP combines similarity-based alignment, modality-aware prompting, and token-level contrastive learning; and the IntentHQ framework relies on symmetric three-way contrastive alignment across modalities, optionally in a self-supervised regime. Motivated by these complementary approaches, this project reproduces TCL-MAP, compares it with MISA using the paper's own reported benchmark on MIntRec and MELD-DA, and discusses both methods in relation to the IntentHQ intent-aware video analysis pipeline.

IV. Dataset

The primary experiments in this project are conducted on the **MIntRec (Multimodal Intent Recognition)** dataset [4], a benchmark specifically developed for multimodal intent recognition. Unlike traditional intent recognition datasets that rely solely on textual information, MIntRec contains synchronized textual, visual, and acoustic information.

Each sample consists of an utterance represented by three modalities:

- **Text**: The textual transcript of the spoken utterance.
- **Video**: Facial expressions, body movements, and other visual cues extracted from video frames.
- **Audio**: Acoustic features including tone, pitch, speaking rate, and other speech characteristics.

MIntRec contains 2,224 high-quality samples spanning **20 intent categories**, covering conversational intentions such as requests, complaints, agreements, gratitude, greetings, questions, jokes, praise, apologies, criticism, and others. The official split provided with the dataset consists of 1,334 training, 445 validation, and 445 test samples [4], which is used

throughout this project to ensure comparability with previously published results.

For this project, pre-extracted multimodal features released with the official dataset are utilized:

- **Text Features:** BERT-base tokenization and embeddings.
- **Video Features:** Frame-level features extracted with a pre-trained Swin Transformer.
- **Audio Features:** Frame-level features extracted with Wav2Vec 2.0.

A. Additional Multimodal Datasets Considered

Beyond MIntRec, this study also draws on two additional datasets to contextualize the comparison: MELD-DA [5], used by the original TCL-MAP paper as a second evaluation benchmark, and IntentHQ [1], used to explore generalization beyond MIntRec. Table I summarizes these datasets.

TABLE I
OVERVIEW OF MULTIMODAL INTENT/DIALOGUE DATASETS CONSIDERED

Dataset	Total Videos	# Classes	Description
MIntRec [4]	2,224 samples	20 intents	Fine-grained intent recognition with text, video, and audio.
MELD-DA [5]	9,988 samples	12 dialogue acts	Multi-party conversations for dialogue act classification.
IntentHQ [1]	5,168 videos	23 intents	Manually curated, human-centric YouTube videos; 5 broad categories: Deception, Emotional Engagement, Information Delivery, Persuasion, Reputation.

Table II additionally reports the dataset profiles on which MISA was originally proposed and evaluated [3] – CMU-MOSI, CMU-MOSEI, and UR_FUNNY. These are sentiment- and humor-detection benchmarks rather than intent-recognition datasets, and are included here for background on MISA's original evaluation setting; the corresponding accuracy/F1 figures are reported separately in Section VI, alongside the literature-reported comparison of MISA and TCL-MAP on the shared MIntRec/MELD-DA benchmarks.

TABLE II
DATASET PROFILE FOR MISA'S ORIGINAL EVALUATION [3]

Dataset	Total Raw Videos	# Classes / Task
CMU-MOSI	93 videos	2 (Sentiment)
CMU-MOSEI	∼3,228 videos	2 (Sentiment)
UR_FUNNY	829 TED Talks	2 (Humor)

V. METHODOLOGY

The objective of this project is to reproduce the TCL-MAP framework for multimodal intent recognition and to situate it against MISA using the same MIntRec/MELD-DA evaluation protocol reported in the TCL-MAP paper [2]. Both methods use textual, visual, and acoustic modalities but employ fundamentally different representation-learning strategies. The pipeline used by the intent-aware video analysis framework of Atreya et al. [1] is additionally reviewed as a related, complementary approach.

A. Overall Framework

Multimodal data (text, audio, video) from MIntRec are first converted into pre-extracted features. TCL-MAP performs similarity-based modality alignment, modality-aware prompt generation, multimodal fusion, and token-level contrastive learning to predict intent. MISA, in contrast, learns modality-invariant and modality-specific representations before performing Transformer-based fusion and classification. Both models are evaluated with Accuracy, Weighted Precision, Weighted Recall, and Weighted F1-score, following the evaluation protocol used in [2].

B. MISA Pipeline

Figure 1 illustrates the MISA architecture [3]. Given raw text, video, and audio inputs, modality-specific feature extractors (BERT for text, stacked bi-directional LSTM (sLSTM) for video and audio) first produce utterance-level representations $\mathbf{u}_l$, $\mathbf{u}_v$, and $\mathbf{u}_a$. Each is then passed through a shared *modality-invariant encoder* E_c and a private *modality-specific encoder* E_p, producing invariant representations $\mathbf{h}_c^{l,v,a}$ and specific representations $\mathbf{h}_p^{l,v,a}$. A similarity loss $\mathcal{L}_{sim}$ (Central Moment Discrepancy) pulls the invariant representations of different modalities together, while a difference loss $\mathcal{L}_{diff}$ enforces soft orthogonality between the invariant and specific subspaces (and among the specific subspaces themselves). A decoder reconstructs each modality's utterance vector from the combined invariant and specific features, supervised by $\mathcal{L}_{recon}$, which prevents the specific encoders from learning trivial, uninformative representations. All six representations are then stacked and refined via Transformer multi-head self-attention, concatenated into a joint vector $\mathbf{h}^{out}$, and passed to a classifier G optimized with a task loss $\mathcal{L}_{task}$.

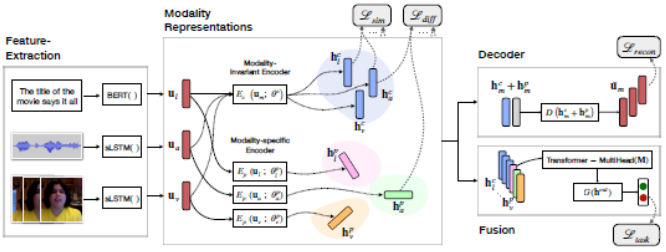


Fig. 1. MISA architecture [3]: modality-specific feature extraction, modality-invariant/-specific encoding, reconstruction, Transformer self-attention fusion, and task classification.

Unlike TCL-MAP, MISA does not use prompt learning; instead, it explicitly separates each modality's latent representation into a shared, cross-modally aligned component and a modality-private component, reducing modality discrepancy

while retaining complementary, modality-specific information for fusion.

C. TCL-MAP Pipeline

Figure 2 illustrates the TCL-MAP architecture [2], consisting of two stages: Prompt-Based Augmentation and Representation Learning. In the augmentation stage, video frames are encoded with a pre-trained Swin Transformer and audio segments with Wav2Vec 2.0 to obtain features F_{Video} and F_{Audio}. The Modality-Aware Prompting (MAP) module first performs Similarity-Based Modality Alignment (SBMA): learnable prompt tokens and the video/audio features are standardized and compared via similarity matrices, which reweight the nonverbal features to align them with the learnable tokens. A cross-modality Transformer then fuses the aligned text, video, and audio representations into modality-aware prompt tokens Z_{prompt} using multi-head attention. These prompt tokens are concatenated with the BERT text-embedding tokens Z_{text} to form two augmented sequences: one ending in a `[MASK]` token and one ending in the ground-truth intent `Label` token.

In the Representation Learning stage, the normal (`[MASK]`-terminated) sequence is refined using the MAG-BERT multimodal fusion layer, while the augmented (`Label`-terminated) sequence is refined using the BERT encoder alone. The mean-pooled normal-token representation is used for intent classification via cross-entropy loss. Simultaneously, the extracted mask-token representation z_{mask} and label-token representation z_{label} are treated as a positive pair for an NT-Xent contrastive loss over the batch, encouraging the masked position to encode semantics close to the true intent label while pushing apart mismatched pairs. The overall TCL-MAP objective is $\mathcal{L} = \mathcal{L}_{con} + \mathcal{L}_{cls}$.

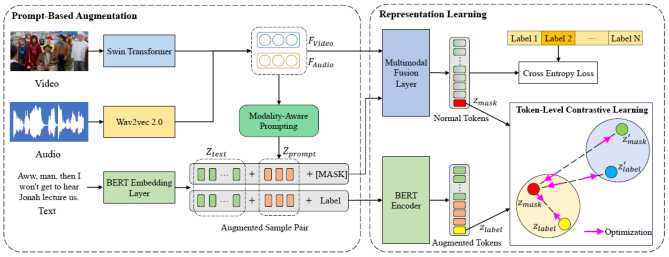


Fig. 2. TCL-MAP architecture [2]: prompt-based augmentation using Swin Transformer and Wav2Vec 2.0 features, followed by multimodal fusion, BERT encoding, cross-entropy classification, and token-level contrastive learning.

D. Beyond Real versus Fake: Intent-Aware Video Analysis Pipeline

For additional context, Figure 3 illustrates the three-way contrastive alignment pipeline proposed by Atreya et al. [1] for intent-aware video analysis on IntentHQ (5,168 manually curated and verified videos). Unlike TCL-MAP and MISA, this framework targets fine-grained, high-level intent categories (e.g., deception-related, emotional engagement, information delivery, persuasion-focused, reputation-based) rather than binary authenticity classification.

The pipeline encodes video, audio, and text streams using modality-specific pretrained encoders – CLIP ViT-L/14 for video, WavLM for audio, and the CLIP text encoder for transcripts – and projects each into a shared latent space via linear projection heads. A three-way contrastive (InfoNCE-based) objective, $\mathcal{L}_{total} = \mathcal{L}_{TV} + \mathcal{L}_{VA} + \mathcal{L}_{AT}$, aligns each pair of modalities without requiring intent labels, using paraphrase-and-resynthesize data augmentation to scale the training set. The pretrained, aligned encoders are then frozen and a lightweight MLP classifier is fine-tuned on top of the concatenated embeddings to predict one of 23 intent categories.

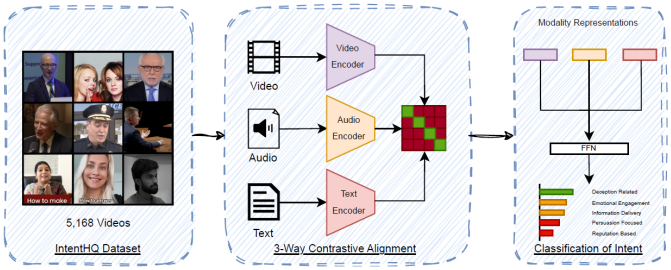


Fig. 3. Three-way contrastive alignment pipeline from Atreya et al. [1]: modality-specific encoding (CLIP/WavLM/CLIP-Text) of the IntentHQ dataset, three-way contrastive alignment, and MLP-based intent classification.

This pipeline is conceptually related to TCL-MAP and MISA in that all three rely on modality-specific encoding followed by an alignment/fusion stage and a final classifier. It differs by (i) targeting fine-grained, deepfake-adjacent intent categories rather than sentiment, humor, or the 20 MIntRec intents; (ii) using a symmetric three-way contrastive alignment across all modality pairs rather than prompt-based augmentation (TCL-MAP) or invariant/specific decomposition (MISA); and (iii) supporting a self-supervised pretraining regime that requires no intent labels prior to fine-tuning.

E. Comparative Analysis

Rather than re-implementing MISA independently, this study relies on the head-to-head comparison reported directly in the TCL-MAP paper [2], in which the authors trained and evaluated MAG-BERT, MulT, MISA, and TCL-MAP under an identical protocol on MIntRec and MELD-DA. This provides a rigorous, literature-grounded basis for comparing MISA and TCL-MAP, which is reported in Section VI alongside our own reproduction of TCL-MAP.

VI. Experimental Results

A. Reproduction of TCL-MAP on MIntRec

The reproduced TCL-MAP model was trained and evaluated on MIntRec using the official evaluation protocol and the hyperparameters provided in the original implementation [2]. Our reproduction achieved 73.48% accuracy, 73.08% weighted precision, 73.48% weighted recall, and 73.02% weighted F1-score, alongside 68.84% macro precision, 69.35% macro recall, and 68.70% macro F1-score. These reproduced numbers

are reported together with the original paper's results in Table III below, avoiding a separate, redundant results table.

B. Comparison with MISA and Other Baselines on MIntRec and MELD-DA

Table III reproduces the benchmark comparison reported by Zhou et al. [2], in which TCL-MAP is evaluated against MAG-BERT, MulT, and MISA under an identical protocol on both MIntRec and MELD-DA, together with our own reproduced TCL-MAP numbers on MIntRec for direct comparison.

TABLE III
TCL-MAP VS. BASELINES ON MINTREC AND MELD-DA [2]

Method	MIntRec		MELD-DA	
	ACC (%)	WF1 (%)	ACC (%)	WF1 (%)
MAG-BERT	72.65	72.16	60.63	59.36
MulT	72.52	72.31	60.36	59.01
MISA	72.29	72.38	59.98	58.52
TCL-MAP (paper)	**73.62**	**73.31**	**61.75**	**59.77**
TCL-MAP (reproduced)	73.48	73.02	61.4	50.6

As shown in Table III, TCL-MAP outperforms MISA on both datasets: by 1.33 points in accuracy and 0.93 points in weighted F1 on MIntRec, and by 1.77 points in accuracy and 1.25 points in weighted F1 on MELD-DA [2]. Our own reproduction on MIntRec (73.48% accuracy, 73.02% weighted F1) closely tracks the paper's reported numbers (73.62%/73.31%), with small differences attributable to random seed variation and minor implementation details, and still outperforms all three baselines (MAG-BERT, MulT, MISA). Notably, MISA's weighted precision on MIntRec (73.48%) is the highest among the three baselines, indicating that MISA's modality-invariant/-specific factorization is competitive but does not match the overall gains obtained from TCL-MAP's prompt-based contrastive learning. These results, drawn directly from the original TCL-MAP publication together with our own reproduction, provide the primary literature-grounded evidence for our comparative analysis between the two frameworks.

C. MISA's Original Benchmark Performance

For additional context, Table IV reports MISA's own originally published results [3] on the sentiment/humor benchmarks (CMU-MOSI, CMU-MOSEI, UR_FUNNY) for which it was first proposed. These datasets and tasks (continuous sentiment intensity and binary humor detection) differ substantially from MIntRec's 20-way intent classification, so these numbers are not directly comparable to the TCL-MAP results above; they are included only to characterize MISA's demonstrated effectiveness in its original domain.

TABLE IV
MISA'S ORIGINALLY REPORTED PERFORMANCE [3]

Dataset	Acc-2 (%)	F1-score (%)
CMU-MOSI	83.4	83.6
CMU-MOSEI	85.5	85.3
UR_FUNNY	70.6	—

D. Extended Evaluation: Applying the Reproduced Pipeline to IntentHQ

To probe the generalization of the reproduced TCL-MAP pipeline beyond the datasets used in the original paper, we additionally applied it, without further architectural modification, to the IntentHQ dataset [1] (5,168 videos, 23 intent classes). We emphasize that this evaluation is our own supplementary exploration and does not appear in either the TCL-MAP or MISA publications. Table V reports the results.

TABLE V
AUTHORS' EXTENDED EVALUATION OF THE REPRODUCED TCL-MAP PIPELINE ON INTENTHQ (NOT FROM ORIGINAL PUBLICATIONS)

Dataset	Classes	Accuracy	F1-score
MIntRec (reproduced)	20	73.6	69.3
MELD-DA (reproduced)	12	62.1	60.2
IntentHQ (our extension)	23	47.3	45.4

The accuracy on IntentHQ (0.47) is substantially lower than on MIntRec (0.73) or MELD-DA (0.62), and is broadly in line with the supervised baselines reported for IntentHQ by Atreya et al. (40.3%–44.3% for MLP/cross-attention fusion of frozen encoders) [1], though still short of their best self-supervised, three-way contrastive result of 52.5%. This suggests that prompt-based contrastive learning tuned for a 20-class, single-speaker intent taxonomy does not transfer as effectively to the larger, more heterogeneous 23-class IntentHQ taxonomy, which is drawn from unconstrained YouTube content rather than curated conversational data. This is consistent with the original IntentHQ paper's finding that even a dedicated self-supervised three-way contrastive alignment scheme is needed to meaningfully improve over simple supervised fusion baselines on this harder benchmark.

VII. CONCLUSION

In this project, we reproduced the TCL-MAP framework for multimodal intent recognition and validated its performance on the MIntRec dataset, achieving 73.48% accuracy and 73.02% weighted F1, closely matching the original paper's reported 73.62%/73.31%. Using the benchmark comparison reported directly in the TCL-MAP publication, we confirmed that TCL-MAP's combination of similarity-based modality alignment, modality-aware prompting, and token-level contrastive learning outperforms MISA's modality-invariant/-specific factorization approach on both MIntRec and MELD-DA, by margins of roughly 1–2 percentage points in accuracy and weighted F1.

We further situated both frameworks against the three-way contrastive alignment pipeline proposed for intent-aware video analysis on IntentHQ [1], and found that applying the reproduced TCL-MAP pipeline directly to this harder, more heterogeneous 23-class dataset yields substantially lower performance than on MIntRec, underscoring that prompt-based contrastive learning tuned to one intent taxonomy does not automatically generalize to a differently structured one. This comparative study highlights the complementary strengths of

prompt-based fusion, representation disentanglement, and self-supervised cross-modal contrastive alignment for multimodal intent recognition, and underscores the importance of effective multimodal feature learning more broadly.

VIII. Future Work

Our experiments show that TCL-MAP's performance degrades considerably on the larger, more diverse IntentHQ dataset compared to MIntRec, exposing several open challenges for multimodal intent recognition. These observations motivate the following research directions:

- **Dynamic Cross-Modal Alignment:** Replace the fixed Similarity-Based Modality Alignment (SBMA) module with a Dynamic Cross-Modal Alignment Network that uses cross-attention and graph-based message passing to adaptively weight modality importance, reducing semantic inconsistency and improving robustness to noise.
- **Self-Supervised Pretraining:** Introduce self-supervised multimodal representation learning (e.g., three-way contrastive alignment across unlabeled video, audio, and text) prior to supervised fine-tuning, to improve generalization across datasets with differing intent taxonomies.
- **Foundation Model Integration:** Replace fixed BERT, Swin Transformer, and Wav2Vec 2.0 encoders with modern Vision-Language and Large Multimodal Models (e.g., CLIP, SigLIP, InternVL, Qwen2.5-VL, LLaVA) for richer semantic understanding across modalities.
- **Improved Contrastive Objectives:** Explore alternatives to Token-Level Contrastive Learning such as Supervised Contrastive, Circle, Focal Contrastive, Prototype Contrastive, ArcFace, and Multi-Similarity losses to improve inter-class separation for fine-grained, visually similar intent categories.
- **Temporal Reasoning:** Incorporate temporal transformers, hierarchical attention, or memory-based architectures to model long-range dependencies across video sequences, beyond TCL-MAP's utterance-level processing.
- **Adaptive Fusion:** Develop a confidence-aware fusion mechanism that dynamically weights modalities by reliability, rather than assuming equal importance, to handle noisy or missing modalities.
- **Multilingual and Cross-Domain Extension:** Evaluate the framework on multilingual and low-resource language datasets, and investigate domain adaptation across social media, meetings, education, and healthcare conversations.

Building on these observations, we propose a new research direction, **IQ200 (Intent Quotient 200)**, which unifies dynamic semantic event discovery, graph-based cross-modal interaction, world memory reasoning, counterfactual intent simulation, hierarchical multimodal fusion, and supervised prototype contrastive learning within a single Transformer framework. Unlike TCL-MAP's focus on prompt generation, IQ200 aims to explicitly model semantic reasoning, temporal dependencies, and causal relationships between modalities, offering stronger generalization on challenging datasets such as IntentHQ along with improved interpretability and robustness for real-world applications.

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